# Safety systems · US & Canada

Each railroad's rule book names the aspects. **NORAC** (Northeast Corridor, NJ Transit, Horseshoe Curve) and **CROR** (Canada) are speed signalling: the aspect names the speed, Limited 45, Medium 30, Slow 15 mph. **GCOR** (BNSF, UP, Metrolink, Caltrain, Tri-Rail) is route signalling: "Diverging" means the turnout at its posted speed. LIRR and Metro-North have their own books with the same aspects. **Restricted speed**: max 20 mph (Canada 15), able to stop within half the range of vision. On the corridor the **cab signal** rules and **ACSES** enforces the limits; the **alerter** checks that you are awake.

## Safety systems

### Cab signals · ATC

Pulse-code cab signals with Automatic Train Control

The cab display repeats the signal aspect and its speed. On a downgrade an alarm sounds: **acknowledge** within about 6 s and make a **suppression** brake application, or ATC applies a penalty brake.

**WHERE** Northeast Corridor, LIRR, Metro-North, NJ Transit, MBTA, Tri-Rail; Sherman Hill (UP, buggy display)

**QUADRANT** **Gear lever** on the Amtrak, LIRR, Metro-North stock and the NJT Multi-Level; **ENG 2** on the MBTA cab cars and the SD70ACe. NJT locos, Arrow III, Tri-Rail locos: in the cab

### ACSES II

Advanced Civil Speed Enforcement System

Transponders give the civil speed limits and **positive stops** at home signals. The display shows the target speed; brake before the restriction. Warning: about 8 s to brake.

**WHERE** Amtrak NEC, NJ Transit, Metro-North, LIRR, MBTA

**QUADRANT** Left aux square = ACSES · ATC fuse or cut-out

### PTC (I-ETMS)

Positive Train Control

In the real world GPS and wayside messages enforce signals, speed limits and authorities. **In TSW not enforced**: the PTC fuse only lights the display.

**WHERE** Real: BNSF, UP, Metrolink, Caltrain (I-ETMS), Tri-Rail. TSW: fuse on the F125, F59PHR, MP36, Arrow III, Tri-Rail locos

**QUADRANT** PTC fuse on the left aux square where fitted

### ATS

Automatic Train Stop (intermittent)

An inductor at each signal showing anything but Clear sounds an alarm: **acknowledge** (forestall) within about 8 s or the brakes apply.

**WHERE** Ex-Santa Fe lines: Metrolink San Bernardino and Antelope Valley; TSW's Tri-Rail Rotem also has it

**QUADRANT** **ENG 2** ATS ack on the Rotem cab cars; F125, F59PHR, MP36: in the cab. Left aux = ATS fuse

### Alerter

Driver vigilance

With no control moved for a while (shorter at speed), a light then a buzzer: **reset** or the train gets a penalty brake. Any control movement also resets.

**WHERE** Every modern locomotive and cab car

**QUADRANT** **Gear lever** = alerter reset on most locos; Caltrain Bi-Level: ENG 2; F7s: safety pedal / deadman. Aux square = alerter fuse

### CROR signals

Canadian Rail Operating Rules

Canada names aspects by this and the next signal: Clear to Stop, Clear to Limited, Limited to Clear. Same lamps, different grammar (flashing green = Clear to Limited).

**WHERE** Oakville Subdivision (CN, Metrolinx)

**QUADRANT** CN locos: gear lever = alerter

## Cab signal aspects (Northeast Corridor, 9-aspect)

| Cab aspect | Max speed | Meaning |
|---|---|---|
| **Clear 150 / 125 / 100** | 150, 125 or 100 mph | Proceed; ACSES enforces the civil limit |
| **Cab Speed 80 / 60** | 80 or 60 mph | Proceed at the cab speed (wayside: flashing green); used for high-speed turnouts |
| **Approach Limited** | 45 mph | Proceed approaching the next signal at Limited speed |
| **Approach Medium** | 30 mph | Proceed approaching the next signal at Medium speed |
| **Approach** | 30 mph | Prepared to stop at the next signal |
| **Restricting** | 20 mph | Restricted speed |

Sources: GCOR (General Code of Operating Rules) · NORAC Operating Rules · Amtrak NORAC / cab signal instructions · CROR (Canadian Rail Operating Rules) · FRA 49 CFR 236 · en.wikipedia: Railroad signal, Cab signaling, ACSES, Positive train control, Alerter



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## Colour-light signals (GCOR, two heads)

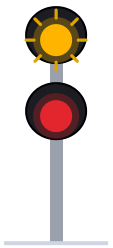
Upper head = the main route, lower head = the diverging route. A number plate makes a signal permissive: Stop and Proceed at restricted speed. Rays = flashing.



### Clear

Proceed at track speed.

**DRIVE** Track speed.



### Advance Approach

Prepare to pass the next signal at 40 mph (UP) or 50 mph (BNSF).

**DRIVE** Start braking; next signal Approach.



### Approach Diverging

Prepare to take the diverging route at the next signal (BNSF: Approach Medium).

**DRIVE** Be at the turnout speed at the next signal.



### Approach

Prepare to stop at the next signal.

**DRIVE** Reduce to 30 mph at once, stop short of the next signal.



### Diverging Clear

Take the diverging route at its posted speed.

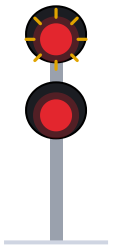
**DRIVE** Turnout speed through the points.



### Diverging Approach

Diverging route, then prepare to stop at the next signal.

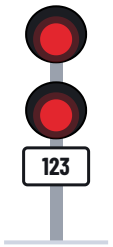
**DRIVE** Turnout speed, then brake.



### Restricting

Proceed at restricted speed.

**DRIVE** Max 20 mph, ready to stop short.




### Stop and Proceed

Number plate: stop, then proceed at restricted speed.

**DRIVE** Stop, then max 20 mph.



### Stop

Absolute signal at an interlocking: stop.

**DRIVE** Stop before the signal; ACSES enforces it on the corridor.

## Position-light signals (Pennsylvania Railroad)

Rows of three lights: the angle of the row is the aspect. Horseshoe Curve keeps the amber originals; on the NEC (New York – Trenton) the same rows are coloured: vertical green, diagonal yellow, horizontal red. Sand Patch has B&O colour-position lights.



### Clear

Vertical row: proceed.

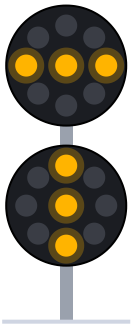
**DRIVE** Track speed.



### Approach

Diagonal row: prepare to stop at the next signal.

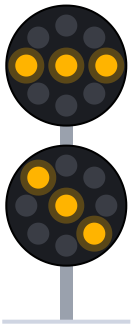
**DRIVE** Brake to stop short of the next signal.



### Medium Clear

Horizontal over vertical: medium speed through the interlocking.

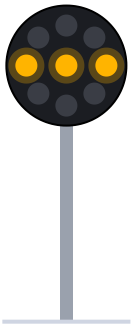
**DRIVE** Max 30 mph through the points.



### Restricting

Horizontal over the falling diagonal: restricted speed.

**DRIVE** Max 20 mph, ready to stop.



### Stop

Horizontal row: stop.

**DRIVE** Stop before the signal.

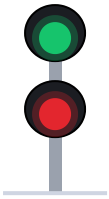
Sources: GCOR (General Code of Operating Rules) · NORAC Operating Rules · Amtrak NORAC / cab signal instructions · CROR (Canadian Rail Operating Rules) · FRA 49 CFR 236 · en.wikipedia: Railroad signal, Cab signaling, ACSES, Positive train control, Alerter



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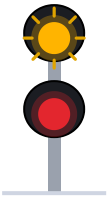
# Safety systems · US & Canada

## CROR signals (Canada)



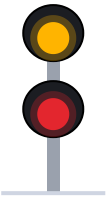
### Clear
Proceed.

**DRIVE** Track speed.



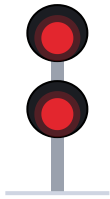
### Advance Clear to Stop
Proceed, preparing to stop at the second signal.

**DRIVE** Next signal Clear to Stop.



### Clear to Stop
Proceed, preparing to stop at the next signal.

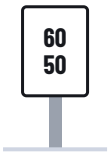
**DRIVE** Brake to stop short of it.



### Stop
Stop.

**DRIVE** Stop before the signal.

## Signs and boards




### Speed sign
Maximum speed from here; two numbers = passenger / freight.

**DRIVE** Lower: by the sign. Higher: once the rear has passed.



### Yellow board
Temporary speed restriction ahead (slow order).

**DRIVE** Brake to the slow order speed.



### Green board
End of the slow order.

**DRIVE** Resume once the rear has passed.




### Whistle post
A grade crossing ahead.

**DRIVE** Horn: long, long, short, long; bell on.



### Red flag / board
Stop: work on the track ahead.

**DRIVE** Stop; proceed only when authorised.

## How to drive under cab signals, ACSES, PTC, ATS and the alerter

### Cab signal downgrade (ATC)
1. The cab display drops (e.g. Clear to Approach) and the alarm sounds.
2. Within about 6 s: **Acknowledge** (see the card for your train) and put the automatic brake in **Suppression** (EMUs: controller to Coast and brake).
3. Release once below the new cab-signal speed.
4. Some trains (LIRR) want every cab-speed change acknowledged, upgrades too.

### ACSES speed and stop enforcement
1. Watch the ACSES target speed and distance on the display.
2. When the warning sounds, acknowledge and brake to the target before the restriction.
3. Positive stop: ACSES stops you within 1000 ft of an absolute signal at Stop. With permission press Stop Release (in the cab), then max 15 mph through the interlocking.

### PTC in TSW
- Keep the PTC fuse in where the train has one; the display may light, but TSW does not enforce PTC.
- Drive to the wayside signals and the speed signs yourself.
- Real I-ETMS counts down to an enforcement; brake and the warning clears.

### ATS (Metrolink, Tri-Rail Rotem)
1. Passing an inductor at any signal but Clear the ATS alarm sounds.
2. Press **ATS acknowledge** within about 8 s: ENG 2 on the Rotem cab cars, in the cab on the F125, F59PHR and MP36.
3. Then drive to the signal aspect.

### Alerter
1. Keep the alerter cut in (aux square, fuse in).
2. When the alerter light flashes or the buzzer sounds, reset it with the **gear lever** (or move a control).
3. Missed: penalty brake. Stop, brake handle to Suppression, reset, then release.

### Penalty brake recovery
1. Throttle to Idle, automatic brake handle to **Suppression** (or Full Service).
2. Wait until the PCS lamp goes out (a minute or so).
3. Release the brake, recharge, then continue.

**Before departure:** Alerter, ATC / ACSES (or PTC, ATS) cut in: fuses in, cut-out squares as the train's manual says. Cab signal and wayside signal agree. Check in the train's manual which control acknowledges the cab signal: gear lever, ENG 2 or the cab.

Sources: GCOR (General Code of Operating Rules) · NORAC Operating Rules · Amtrak NORAC / cab signal instructions · CROR (Canadian Rail Operating Rules) · FRA 49 CFR 236 · en.wikipedia: Railroad signal, Cab signaling, ACSES, Positive train control, Alerter



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